

Ye Wang

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🌐 LinkedIn | 🐙 Github | 🎓 Google Scholar | 🗂 ORCID

RESEARCH INTEREST

My research primarily focuses on Physical AI systems and their security. Specifically, I investigate both offensive and defensive techniques for IoT, CPS, and AI-enabled systems. My research interests include:

- Trustworthy and Privacy-preserving ML/AI: Adversarial ML; Prompt Injection; AI Misinformation and Misuse.
- Physical AI: IoT/CPS; Mobile; Wireless Systems; Side and Covert Channels.
- Communication Security: NFC; Optical Fiber Communication; GPS Spoofing.

EDUCATION

- **University of Kansas** Lawrence, Kansas, United States
Ph.D. in Computer Science, Department of Electrical Engineering and Computer Science Jan. 2020 – Mar. 2026
Advisor: [Prof. Fengjun Li](#) & [Prof. Bo Luo](#)
- **Beihang University** Beijing, China
M.Eng. in Optical Engineering, School of Instrumentation Science and Optoelectronic Engineering Sep. 2011 – Mar. 2014
Advisor: [Prof. Xiaoxiao Wang](#)
- **Beihang University** Beijing, China
B.S. in Electronic Engineering, School of Instrumentation Science and Optoelectronic Engineering Sep. 2007 – Jul. 2011
Advisor: [Prof. Zhongyi Chu](#)

PUBLICATIONS

Students mentored by me

Conference Papers

- [C1] Ren, Liangqin and Liu, Zeyan and **Wang, Ye** and Chen, Yuxin and Li, Fengjun and Luo, Bo. PhantomSeal: Proactive Deepfakes Defense with Identity/Context Protection and Forensic Tracing. *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 2026. Accepted
- [C2] **Wang, Ye** and Li, Yuying and Luo, Bo and Li, Fengjun. When Side Channels Meet Covert Channels: A Practical Fully Sensor-Driven Attack Chain. *International Conference on Information and Communications Security (ICICS)* 2026. Accepted
- [C3] **Wang, Ye** and Luo, Bo and Li, Fengjun. [Beyond Conventional Triggers: Auto-Contextualized Covert Triggers for Android Logic Bombs](#). *The Network and Distributed System Security (NDSS) Symposium* 2026.
- [C4] **Wang, Ye** and Luo, Bo and Li, Fengjun. [A Novel Fully Sensor-driven Attack Chain](#). *Annual Computer Security Applications Conference (ACSAC)* 2025, Poster Session
- [C5] **Wang, Ye** and Liu, Zeyan and Luo, Bo and Hui, Rongqing and Li, Fengjun. [The Invisible Polyjuice Potion: an Effective Physical Adversarial Attack against Face Recognition](#). *Proceedings of the 2024 ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 2024, 3346–3360.
- [C6] [Li, Kevin](#) and Wang, Zhaohui and **Wang, Ye** and Luo, Bo and Li, Fengjun. [Poster: ethics of computer security and privacy research-trends and standards from a data perspective](#). *Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security (CCS)*, Poster Session 2023, 3558–3560.
- [C7] Fan, Wei and Huang, Weiqing and Zhang, Zhujun and **Wang, Ye** and Sun, Degang. [A Near Field Communication \(NFC\) security model based on the OSI reference model](#). *2015 IEEE Trustcom/BigDataSE/ISPA*, 2015, 1324–1328.

Journal Papers

- [J1] Kong, Qingshan and Kang, Di and **Wang, Ye** and Zhang, Meng and Huang, Weiqing. [Eavesdropping Attacks on Optical Fiber Communication and Countermeasure of Optical Fiber Sensing Technology](#). *Journal of Information Security Research*, 2.2 (2016): 123.
- [J2] Wang, Xiaoxiao and **Wang, Ye*** and Qin, Yi and Yu, Jia. [Ratio error of all fiber optical current transformer caused by mean wavelength's fluctuation](#). *Infrared and Laser Engineering*, 2015, 44.1 (2015): 233-238.
- [J3] Wang, Xiaoxiao and Wang, Xichen and **Wang, Ye** and Feng, Xiujian. [A novel Faraday effect-based semi-physical simulation method for bandwidth of fiber-optic gyroscope](#). *Optik*, 2014, 1358–1360.
- [J4] Wang, Xiaoxiao and **Wang, Ye*** and Wang, Xichen and Wang, Aimin and Peng, Zhiqiang. Experimental research on the dynamic characteristics of fiber-optical current transformer. *Power System Protection and Control*, 42.3 (2014): 9-14.
- [J5] Wang, Xiaoxiao and Qin, Yi and **Wang, Ye***. [Errors of fiber delay line polarization crosstalk for all fiber optical current sensors](#). *Optics and Precision Engineering*, 22.11 (2014): 2930-2936.
- [J6] Wang, Xiaoxiao and **Wang, Ye*** and Li, Chuansheng and others. Measurement method and experimental research of the temperature dependence of the phase delay of quarter-wave plates. *Chinese J Lasers*, 40.12 (2013): 1205004.

RESEARCH AND PROFESSIONAL EXPERIENCE

• Institute for Information Sciences, University of Kansas

Lawrence, Kansas, United States

Jan. 2020 – May. 2026

Graduate Research Assistant

- Non-intrusive physical-layer masking for preventing side-channel leaks via accelerometers.
- Combining motion-sensor side channels with covert vibration channels to form a practical attack chain.
- Stealthy sensor-enabled logic bombs for Android that evade static analysis, dynamic analysis, and user awareness.
- Proactive deepfakes face swap defense with identity/context protection and forensic tracing.
- Develop an effective physical adversarial attack against face recognition CNN models.

• Institute of Information Engineering, Chinese Academy of Sciences

Beijing, China

Mar. 2014 – Dec. 2019

Assistant Research Fellow (full-time)

- Designed and implemented GPS spoofing detection techniques for UAV navigation.
- Developed a fiber-optic communication system security monitoring framework, focusing on intrusion detection.
- Developed an unauthorized recording device detection system based on weak magnetic signal analysis.
- Conducted research on security modeling for Near-Field Communication (NFC) devices.

• Institute of Optoelectronics Technology, Beihang University

Beijing, China

Sep. 2011 – Mar. 2014

Graduate Research Assistant

- Conducted research to improve the dynamic response and measurement accuracy of fiber-optic current transformers.

HONORS AND AWARDS

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| • NDSS Fellowship – Internet Society | 2026 |
| • Doctoral Student Research Fund Award – University of Kansas (KU) | 2025 |
| • Graduate Engineering Association Award – KU | 2024, 2025 |
| • DAVID D. and MILDRED H. ROBB AWARD – EECS, KU | 2024, 2025 |
| • ACM CCS Travel Grant Award – ACM SIGSAC, NSF | 2024 |
| • Graduate Student Travel Fund – KU Student Senate | 2024, 2025 |
| • CANSec Workshop Travel Grant Award – CANSec, NSF | 2022, 2024, 2025 |
| • The second prize of the Science and Technology Award – Ministry of Industry and Information Technology, PRC | 2019 |
| • Annual Outstanding Researcher Award – Institute of Information Engineering, CAS | 2016, 2018 |
| • Outstanding Master's Thesis Award – Beihang University | 2014 |
| • Science and Technology Award – Ministry of Education, PRC | 2013 |
| • Graduate Guanghua Scholarship – Beihang University | 2013 |
| • Outstanding Undergraduate Thesis Award – Beihang University | 2011 |

TEACHING EXPERIENCE

• Graduate Teaching Assistant

University of Kansas

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| ◦ EECS 569: Computer Forensics | Fall 2024 |
| Instructor: Dr. Bo Luo | |
| ◦ EECS 565: Introduction to Information and Computer Security | Fall 2025, 2024, 2023 |
| Instructor: Dr. Fengjun Li | |
| ◦ EECS 447: Introduction to Database Systems | Spring 2026, 2024, 2023 |
| Instructor: Dr. Bo Luo | |
| ◦ EECS 268: Programming II | Spring 2025 |
| Instructor: Dr. John Gibbons | |

• Teaching Assistant

University of Chinese Academy of Sciences

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| ◦ Physical Space Information Security | Spring, 2017 |
| Instructor: Prof. Degang Sun | |

MENTORING EXPERIENCE

1. Liangqin Ren, PhD candidate, The University of Kansas, 01/2025-present
Project: Proactive Deepfakes Defense with Identity/Context Protection and Forensic Tracing, CCS 2026
2. Yuying Li, PhD student, The University of Kansas, 01/2025-present
Project: A novel attack chain to improve the practicality of side channel attacks, ICICS 2026
3. Weihang Hu, master student, University of Chinese Academy of Sciences, 9/2018-9/2019
Project: CNN-based electromagnetic spectrum analysis (Master's thesis)

CONFERENCE PRESENTATION

- Beyond Conventional Triggers: Auto-Contextualized Covert Triggers for Android Logic Bombs, the Network and Distributed System Security (NDSS) Symposium 2026, February 24th, 2026, San Diego, CA, USA
- A Novel Fully Sensor-driven Attack Chain, Annual Computer Security Applications Conference (ACSAC) poster session, December 11th, 2025, Honolulu, Hawaii, USA
- Invited talk: A novel fully sensor-driven attack chain, I2S Student Organization (ISO) Meeting, October 23rd, 2025, Nichols Hall, The University of Kansas.
- Agile: An Effective Laser-Based Physical Adversarial Attack against Face Recognition, AI summit *Falling into AI* with Google, October 1st, 2025, Nichols Hall, The University of Kansas.
- The Invisible Polyjuice Potion: An Effective Physical Adversarial Attack against Face Recognition, ACM SIGSAC Conference on Computer and Communications Security (CCS '24), October 17th, 2024, Salt Lake City.
- The Invisible Polyjuice Potion: An Effective Physical Adversarial Attack against Face Recognition, The Central Area Networking and Security Workshop (CANSec) 2024, October 12th, 2024, University of Oklahoma, Norman, OK.
- Stealthily Evading Surveillance Camera Face Recognition, FBI and KU Cybersecurity Conference, April 4th, 2024, KU Memorial Union, the University of Kansas.
- Laser Man against Unauthorized Facial Recognition Systems, The I2S Student Research Symposium (ISRS), March 3rd, 2023, Nichols Hall, The University of Kansas.

PROFESSIONAL MEMBERSHIPS

- ACM SIGSAC membership
- Internet Society membership

PROFESSIONAL SERVICE

Paper Review

- External paper reviewer: USENIX Security '26
- Journal reviewer for Neural Networks
- Journal reviewer for IEEE Transactions on Dependable and Secure Computing (TDSC)
- External paper reviewer: ACM SIGSAC Conference on Computer and Communications Security (CCS 2025)
- External paper reviewer: the 44th IEEE International Conference on Distributed Computing Systems (ICDCS 2024)
- External paper reviewer: Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN 2023)
- External paper reviewer: the Annual Computer Security Applications Conference (ACSAC 2023)
- External paper reviewer: Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN 2022)
- External paper reviewer: IEEE International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom 2022)

Community Service

- GenCyber Teacher Camp, Student Volunteer and Teaching Assistant, Funded by NSA and NSF, 2023
- GEA Research Symposium, Student Judge, University of Kansas, 2023
- Session Moderator for EAI SecureComm, 2022

REFERENCES

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| • Dr. Fenjun Li
fli@ku.edu | <i>EECS, University of Kansas</i>
Deane E. Ackers Professor |
| • Dr. Bo Luo
bluo@ku.edu | <i>EECS, University of Kansas</i>
H.J. and Joan O. Wertz Professor |
| • Dr. Rongqing Hui
rhui@ku.edu | <i>EECS, University of Kansas</i>
Professor, Fellow of Optica Society |

Last Update: 5/2026